ϕ field $\Rightarrow$ operator $\hat{\phi}(x,0) \propto \int \frac{d^3k}{\sqrt{\omega_k}} \left(a(k)\, e^{-ik\cdot x} + a^\dagger(k)\, e^{ik\cdot x} \right)$

assumed free Lagrangian

$\langle f | i \rangle$

$$V(\phi) = g \sin(\phi)$$

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2\right) - V(\phi) \rightarrow \partial_\mu \partial^\mu \phi + m^2 \phi = -V'(\phi)$$

$$V(\Psi, \bar{\Psi}, A) = ie\, \bar{\Psi} \gamma^\mu A_\mu \Psi$$

$$\phi(x,t) \propto \int \frac{d^3k}{\sqrt{\omega_k}} \left(a(k)\, e^{i\omega_k t} e^{-ik\cdot x} + a^\dagger(k)\, e^{-i\omega_k t} e^{ik\cdot x} \right)$$ for free field

$$H = \int d^3x \left(\dot{\phi}\Pi - \mathcal{L} \right) = \underbrace{H_0}_{\text{free}} + \underbrace{H_I}_{\text{interaction}} \qquad H_I = \int d^3x\, V(\phi(x))$$

- Heisenberg : $\phi_H(x,t) = e^{+iHt} \phi(x,0)\, e^{-iHt}$
- Interaction : $\phi_I(x,t) = e^{iH_0 t} \phi(x,0)\, e^{-iH_0 t}$

<u>Why?</u> scattering matrix : $|i\rangle$ $|f\rangle$ state asymptotically free

$$S_{fi} = \lim_{t\to\infty} \langle f | i(t) \rangle \qquad |i(t)\rangle = U(t,-\infty)\, |i\rangle$$

$$U(t,t_0) = T e^{-i\int_{t_0}^{t} H_I(\tau)\, d\tau}$$

$\downarrow$ Dyson series

$$\sum_{n=0}^{\infty} \frac{(-i)^n}{n!} T\left(\int d\tau\, H_I \right)^n$$

$$S_{fi} = \lim_{\substack{t_1 \to -\infty \\ t_2 \to +\infty}} \langle f | U(t_2,t_1) | i \rangle$$

$$S_{fi} = \langle f | S | i \rangle \qquad S = \lim_{\substack{t_1 \to -\infty \\ t_2 \to +\infty}} U(t_2,t_1)$$

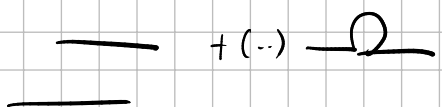
$|i\rangle = \hat{\phi}(x) |0\rangle$

$|f\rangle = \hat{\phi}(x') |0\rangle$

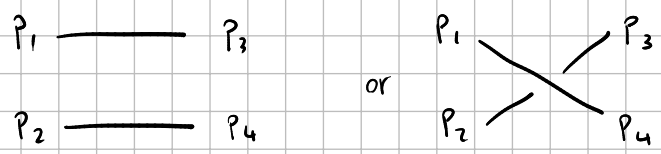
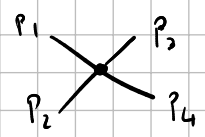
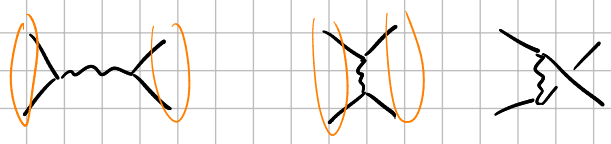
$$\lim_{t\to\infty} \langle f | i(t) \rangle = \langle 0 | \hat{\phi}(x')\, U(t,-\infty)\, \hat{\phi}(x) | 0 \rangle$$

Stop at order 1 $\qquad V(\phi) = g\phi^4 \qquad U(-\infty,+\infty) \propto \mathbb{1} + \int d^4x\, g\hat{\phi}^4(x)$

$$\langle 0|\hat{\phi}(x')\phi(x)|0\rangle - ig\int d^4y\, \langle 0|\hat{\phi}(x')\phi^4(y)\phi(x)|0\rangle$$



2 in, 2 out free theory:



with interactions of order 1: